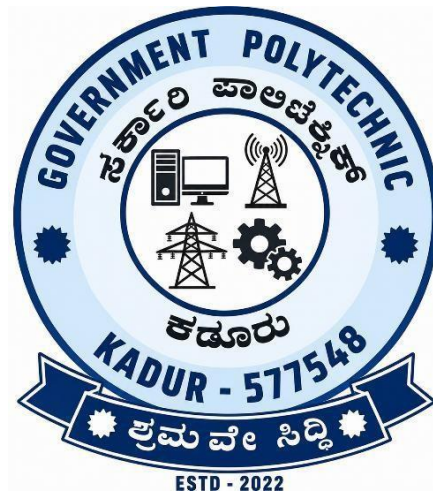


# GOVERNMENT POLYTECHNIC, KADUR

## COMPUTER SCIENCE AND ENGINEERING

### ONLINE COURSE REPORT ON:

PYHTON FUNDAMENTALS



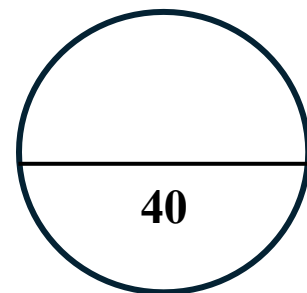
SUBMITTED BY : VASUDEVAN

OBTAINED MARKS:

REGISTER NO : 197CS24028

SEMISTER : 5<sup>th</sup> SEM 3<sup>rd</sup> YEAR

SUBJECT : FULL STACK DEVELOPMENT



**Signature:**

## **INFOSYS SPRINGBOARD**

Infosys Springboard is a free digital learning platform developed by Infosys to provide quality education and industry-relevant skills to students, teachers, and professionals. The platform offers self-paced learning through video lectures, reading materials, practical exercises, and assessments. It helps learners enhance their technical knowledge and prepare for future careers.

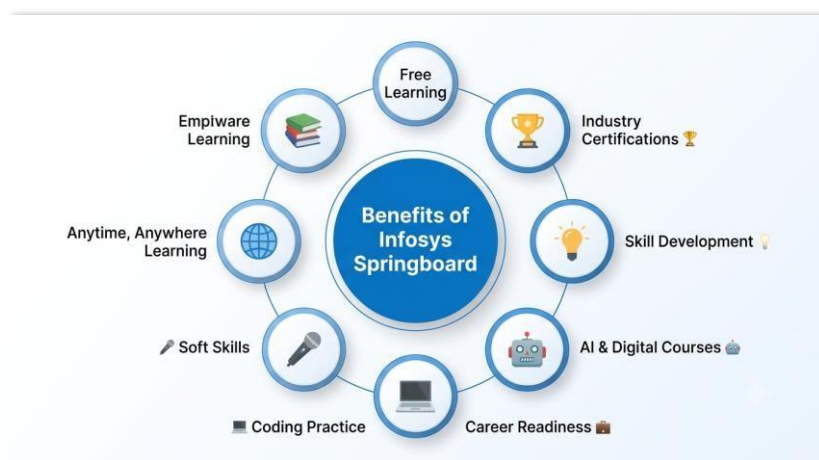


### **Key Features:**

- Free access to a wide range of courses.
- Self-paced online learning.
- Video lectures and study materials.
- Quizzes and assessments.
- Digital certificates after course completion.
- Industry-focused learning paths.
- Progress tracking to monitor learning achievements.
- Mobile-friendly access for learning anytime, anywhere.

## Benefits Of Infosys Springboard:

One of the popular courses on the platform is Introduction to Artificial Intelligence, which covers AI fundamentals, Machine Learning, Deep Learning, AI business applications, and AI platforms such as Infosys NIA. The course helps learners understand how AI is transforming industries like healthcare, banking, retail, manufacturing, and education.



## Improves technical and professional skills:

- Enhances problem-solving and analytical abilities.
- Supports career development and job readiness.
- Provides learning opportunities for students and working professionals.
- Encourages continuous learning and innovation.
- Improves communication and teamwork skills through collaboration on projects and professional interactions.
- Builds confidence in using modern tools and technologies relevant to the industry.

| SL NO | COURSE NAME                             | TIME   |
|-------|-----------------------------------------|--------|
| 1     | INTRODUCTION OF ARTIFICIAL INTELLIGENCE | 6H 18M |
| 2     | STATISTICS ESSENTIALS WITH PYTHON       | 4H 33M |



# ARTIFICIAL INTELLIGENCE



## **Introduction to Artificial Intelligence:**

Artificial Intelligence (AI) is the branch of computer science that develops systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, language understanding, decision-making and pattern recognition. AI improves productivity, automates repetitive work and helps organizations make better decisions.

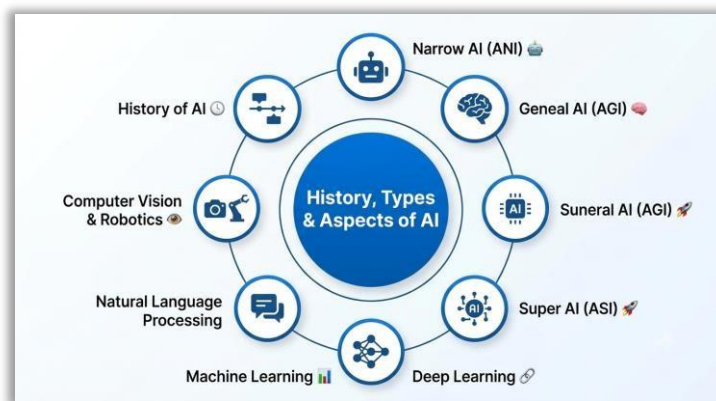
## **Learning Outcomes:**

- Understand AI concepts.
- Identify AI applications.
- Explain machine learning basics.
- Recognize business value of AI.

## Why AI?

AI increases efficiency, reduces errors, supports faster decisions, improves customer experience and enables innovation across industries.

## History, Types and Aspects of AI:



AI began in the 1950s with the Dartmouth Conference. Important milestones include expert systems, machine learning, deep learning and modern generative AI.

- Types of AI: Narrow AI, General AI (theoretical), and Super AI (hypothetical).
- Technical View: AI combines algorithms, data, computing power and models.
- Aspects of AI include perception, reasoning, planning, learning and natural language processing.

## Machine Learning:

- Machine Learning is a subset of AI where computers learn from data.
- Supervised Learning uses labeled data for classification and regression
- Classification predicts categories while regression predicts numerical values.



## Common algorithms:

- Decision Tree
- Support Vector Machine (SVM)
- K-Nearest Neighbours (KNN)
- Logistic Regression

Deep Learning uses Artificial Neural Networks (ANN) with multiple layers for image, speech and text processing.

## Advanced AI Concepts:

- Overfitting occurs when a model memorizes training data, while underfitting occurs when it fails to learn important patterns. Techniques like regularization and cross-validation help improve performance.
- Ensemble Learning combines multiple models, such as Random Forest, to improve prediction accuracy and reliability.
- Unsupervised Learning discovers hidden patterns and groups in unlabelled data using techniques like clustering.
- Reinforcement Learning enables agents to learn through rewards and penalties, making it useful for robotics, gaming, and autonomous systems.



- Time Series Forecasting predicts future values based on historical data and is widely used in finance, sales, and weather forecasting.
- Machine Learning Process includes data collection, preprocessing, feature engineering, model training, evaluation, deployment, and continuous monitoring.
- Model Evaluation uses metrics such as accuracy, precision, recall, and F1-score to measure performance.
- Feature Engineering improves model accuracy by selecting and transforming relevant data features.

## AI in Business:

### **AI in Retail:**

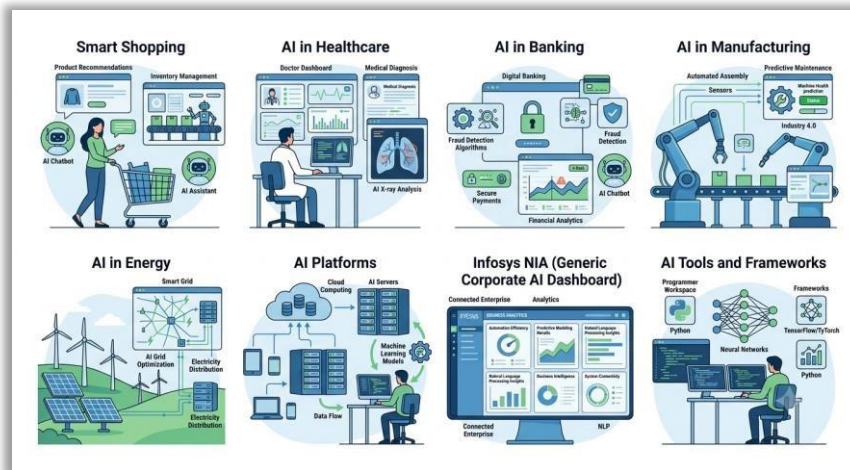
Artificial Intelligence is widely used in the retail industry to improve customer experience and business operations. It helps recommend products based on customer preferences, manages inventory, predicts demand, and provides customer support through AI-powered chatbots. These applications increase sales and improve customer satisfaction.

### **AI in Healthcare:**

AI is transforming healthcare by helping doctors diagnose diseases, analyse medical images, monitor patients, and discover new medicines. It improves the accuracy of treatments, reduces human errors, and enhances the quality of healthcare services.

### **AI in Banking:**

Banks use AI for fraud detection, credit scoring, customer support, and financial risk analysis. AI-powered chatbots provide instant customer assistance, while intelligent systems help make faster and more secure financial decisions.



## AI in Manufacturing:

AI improves manufacturing by automating production processes, predicting machine failures, and inspecting product quality. It helps industries reduce downtime, increase productivity, and lower operational costs.

## AI in Energy:

In the energy sector, AI is used to forecast electricity demand, manage smart grids, optimize renewable energy resources, and detect equipment faults. This helps improve energy efficiency and reliability.

## AI Platforms:

AI platforms provide the tools needed to develop, train, and deploy Artificial Intelligence applications. Popular platforms include Google Vertex AI, Microsoft Azure AI, Amazon SageMaker, IBM Watson, and Infosys NIA. These platforms simplify AI development and deployment.

## Infosys NIA:

Infosys NIA is an enterprise Artificial Intelligence platform developed by Infosys. It combines Machine Learning, Natural Language Processing (NLP), automation, and data analytics to help businesses improve efficiency and make intelligent decisions. It is widely used in industries such as banking, healthcare, retail, and manufacturing.

## **AI Tools and Frameworks:**

Developers use AI tools and frameworks such as TensorFlow, PyTorch, Scikit-learn, Keras, OpenCV, Pandas, and NumPy to build and train AI models. These tools simplify the development of machine learning and deep learning applications.

## **What I Learned:**

During this course, I gained a strong understanding of the fundamentals of Artificial Intelligence and its importance in modern technology. I learned how AI enables machines to perform intelligent tasks and how it is transforming various industries through automation and data-driven decision-making.

## **Key Learnings:**

- Understood the basic concepts and history of Artificial Intelligence.
- Learned about the different types of AI and their applications.
- Gained knowledge of Machine Learning, Deep Learning, and Artificial Neural Networks (ANN).
- Studied Supervised, Unsupervised, and Reinforcement Learning techniques.
- Learned about AI algorithms such as Decision Tree, SVM, KNN, and Logistic Regression.
- Understood AI applications in retail, healthcare, banking, manufacturing, and energy.
- Explored AI platforms like Infosys NIA and tools such as TensorFlow, PyTorch, and Scikit-learn.

This course helped me understand how Artificial Intelligence is used to solve real-world problems, improve business processes, and create innovative solutions. It has also provided me with a strong foundation for learning advanced AI technologies in the future.

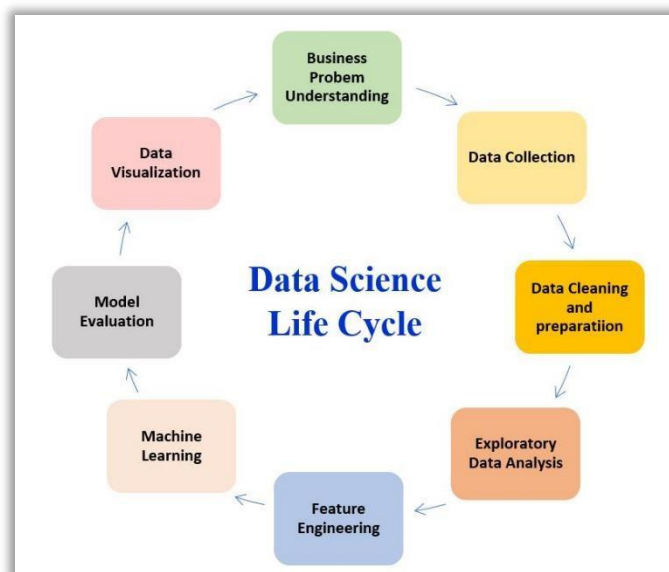
## **Conclusion**

Artificial Intelligence is one of the fastest-growing technologies in the world. It has applications in healthcare, banking, retail, manufacturing, energy, and many other sectors. Learning AI concepts and technologies prepares students for future careers and helps organizations achieve digital transformation.

## **STATISTICS ESSENTIALS WITH PYTHON**

### **Introduction to Data Science**

Data Science collects, processes, analyses, and interprets data to discover meaningful insights. It combines statistics, programming, machine learning, and data visualization to solve real-world problems and support data-driven decision-making, and is widely used across healthcare, finance, education, and business.



- Collects and analyses large amounts of data
- Uses programming languages such as Python and R
- Combines statistics, machine learning, and visualization
- Helps organizations make better decisions
- Identifies patterns and predicts future trends

### **Basic Concept of Statistics:**

Statistics is the branch of mathematics focused on collecting, organizing, analyzing, interpreting, and presenting data. It helps summarize information, identify patterns, and support accurate decision-making, and is fundamental to Data Science, research, and business analytics.

- Used to collect and analyses data
- Divided into Descriptive Statistics and Inferential Statistics
- Measures of central tendency: Mean, Median, and Mode
- Measures of dispersion: Range, Variance, and Standard Deviation
- Helps identify trends, patterns, and relationships in data

## Calculating Mode

Mode is a measure of central tendency representing the value occurring most frequently in a dataset. It applies to numerical or categorical data, and a dataset may be unimodal, bimodal, multimodal, or have no mode at all.

- Mode is the value with the highest frequency
- Can be used for both numerical and categorical data
- A dataset can be unimodal, bimodal, or multimodal
- If no value repeats, the dataset has no mode
- Simple to calculate and interpret

### FORMULA:

***Mode = Value with the highest frequency***

## Calculating Dispersion Measures

Dispersion measures indicate how spread-out values are around the central value, helping determine consistency and variability. Lower dispersion means values are close together; higher dispersion means values are widely distributed.

### FORMULA:

|                          |                                                        |                                                 |
|--------------------------|--------------------------------------------------------|-------------------------------------------------|
| <b>Range = Max – Min</b> | <b>Variance = Avg. of squared deviations from mean</b> | <b>SD = <math>\sqrt{\text{Variance}}</math></b> |
|--------------------------|--------------------------------------------------------|-------------------------------------------------|

## Histogram Calculation

A histogram is a graph showing the frequency distribution of numerical data, grouping values into bins to show how many fall within each interval, revealing the shape, spread, and distribution of a dataset.

- Displays the frequency distribution of data
- Groups values into intervals (bins)
- Helps identify patterns and trends
- Useful for detecting outliers and skewness

**Formula:**

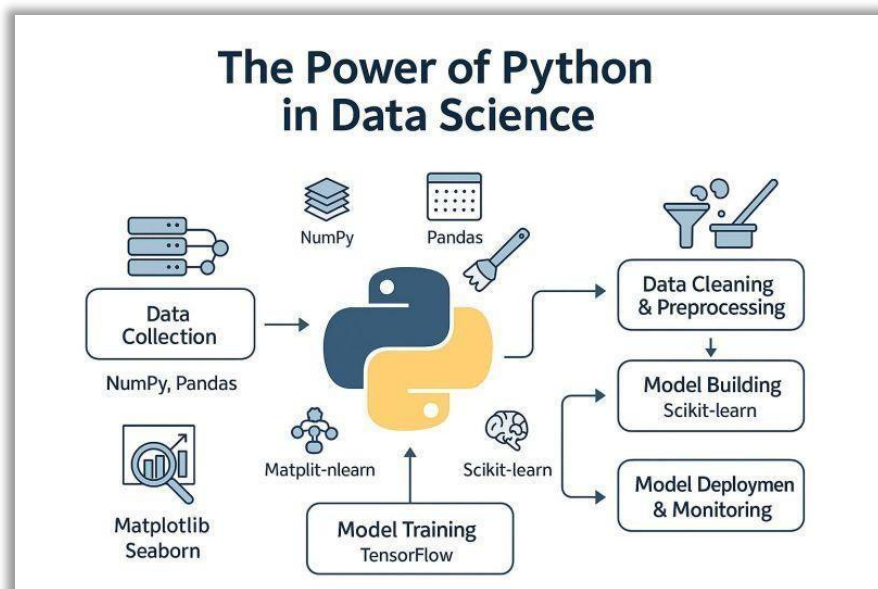
$$\text{Frequency} = \text{Number of observations in each bin}$$

## Correlation Function

A correlation function measures the strength and direction of the relationship between two variables. The coefficient ranges from -1 to +1, where +1 is a perfect positive relationship, -1 a perfect negative relationship, and 0 no relationship.

**Formula:**

$$r = \text{Cov}(X, Y) / (\sigma X \times \sigma Y)$$



## Pandas Data Frame

A Pandas Data Frame is a two-dimensional data structure in Python used to store and organize data in rows and columns. It is one of the most widely used tools in Data Science for cleaning, analysis, filtering, sorting, and visualization.

### Common Operations:

- Creating a Data Frame
- Reading data from CSV or Excel files
- Filtering and sorting records
- Adding or removing columns

## Hypothesis Testing

Hypothesis Testing determines whether a claim about a population is supported by sample data, testing the Null Hypothesis ( $H_0$ ) against the Alternative Hypothesis ( $H_1$ ) using statistical evidence.

## NumPy Package

NumPy (Numerical Python) is a Python library for numerical computing and scientific calculations, supporting multi-dimensional arrays and fast mathematical operations. It performs computations much faster than standard Python lists.

## Finding the Measurement

Finding the measurement is the process of calculating key statistical values from a dataset to understand its characteristics, summarize data, and support accurate analysis.\

## Calculating the Denominator

The denominator is the bottom part of a fraction, representing the total number of values or observations. It is used in formulas for mean, variance, probability, and other statistical measures.

### Formula:

$$\text{Mean} = \Sigma X / n \quad \text{Probability} = \text{Favorable outcomes} / \text{Total outcomes}$$

## Implementing Test Scenarios

Implementing test scenarios involves designing and executing test cases to verify whether statistical methods or models produce accurate, reliable results.

### Formula:

$$P(A \text{ or } B) = P(A) + P(B) \quad [\text{when } A \text{ and } B \text{ are mutually exclusive}]$$

## Ordinary Least Squares (OLS) Regression

OLS Regression determines the relationship between a dependent variable and one or more independent variables by finding the best-fit line that minimizes the sum of squared differences between actual and predicted values.

### Formula:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

### What You Will Learn

- Introduction to Data Science and its applications.
- Basics of Statistics and data analysis.
- Mean, Median, Mode, and measures of dispersion.
- Creating and understanding Histograms.



- Measuring relationships using Correlation.
- Working with Pandas Data Frames and NumPy.
- Basics of Hypothesis Testing and Probability.
- Understanding OLS Regression for prediction.
- Applying Python for real-world data analysis.

## **Conclusion:**

Data Science combines statistics, programming, and machine learning to extract meaningful insights from data. Techniques such as descriptive statistics, hypothesis testing, correlation, regression analysis, and Python libraries like NumPy and Pandas help analyze data efficiently. These methods support accurate predictions, improve decision-making, and solve real-world problems in fields such as healthcare, finance, education, and business. As the volume of data continues to grow, Data Science remains an essential tool for innovation and informed decision-making.